

1. In the figure, two particles, each with mass $m = 0.85$ kg, are fastened to each other, and to a rotation axis at O , by two thin rods, each with length $d = 5.6$ cm and mass $M = 1.2$ kg. The combination rotates around the rotation axis with the angular speed $\omega = 0.30$ rad/s. Measured about O , determine the combination's

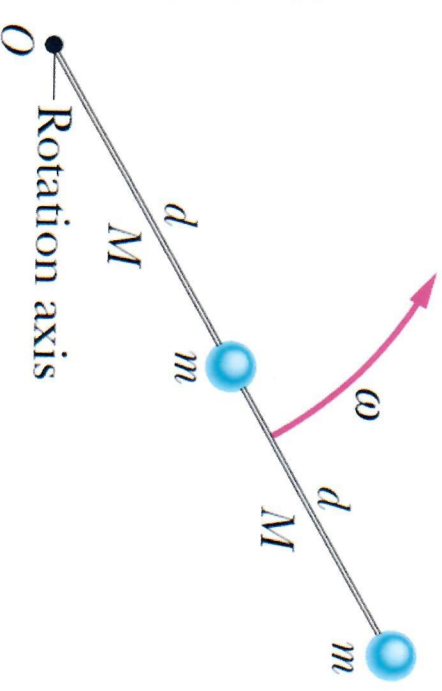
(a) rotational inertia and

(b) kinetic energy.

Die figuur dui aan twee deeltjies, elk met 'n massa $m = 0.85$ kg. Hulle word aan mekaar vasgemaak, en aan die rotasie as vasgemaak deur twee dun stafies, elk met 'n lengte $d = 5.6$ cm en massa $M = 1.2$ kg. Die stelsel roteer om die rotasie-as met 'n hoeksnelheid van $\omega = 0.30$ rad/s. Gemeet om O , bereken die stelsel se

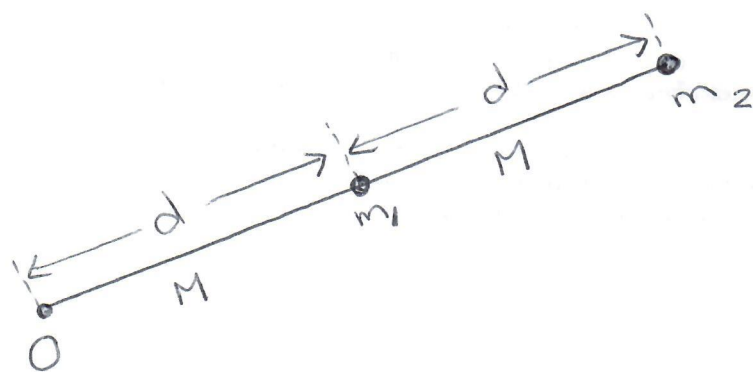
(a) traagheidsmoment en

(b) kinetiese energie.



(10)

Tutorial 10 - 1



- a) The particles can be treated as point masses and the connections as thin rods.

$$\begin{aligned} I &= I_{\text{rod1}} + I_{\text{rod2}} + I_{m_1} + I_{m_2} \\ &= \left(\frac{1}{12} M d^2 + M \left(\frac{d}{2} \right)^2 \right) + \left(\frac{1}{12} M d^2 + M \left(\frac{3}{2} d \right)^2 \right) + \\ &\quad m_1 d^2 + m_2 (2d)^2 \end{aligned}$$

$$= \frac{8}{3} M d^2 + 5 m d^2$$

$$= \frac{8}{3} (1.2 \text{ kg}) (5.6 \text{ m})^2 + 5 (0.85 \text{ kg}) (5.6 \text{ m})^2$$

$$= 234 \text{ kg} \cdot \text{m}^2$$

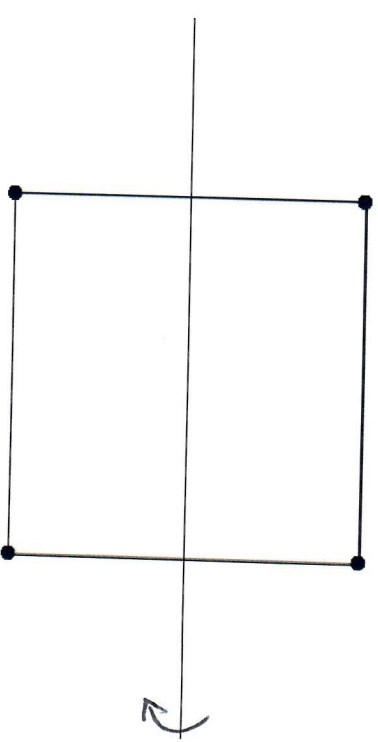
b) $K = \frac{1}{2} I \omega^2$

$$= \frac{1}{2} (0.023 \text{ kg} \cdot \text{m}^2) (0.30 \text{ rad/s})^2$$

$$= 10.5 \text{ J}$$

1. Four identical particles of mass 0.50 kg each are placed at the vertices of a $2.0\text{ m} \times 2.0\text{ m}$ square and held there by four massless rods, which form the sides of the square. What is the rotational inertia of this rigid body about an axis that passes through the midpoints of opposite sides and lies in the plane of the square (See Figure)?

Vier identiese deeltjies elk met massa 0.50 kg word op die hoeke van 'n $2.0\text{ m} \times 2.0\text{ m}$ vierkant geplaas. Hulle word daar in plek gehou deur vier massalose stawe wat die sykante van die vierkant vorm. Wat is die traagheidsmoment van die rigiede liggaam om 'n as wat deur die middelpunt van teenoorstaande sye in die vlak van die vierkant lê (Sien figuur)?



2. Trucks can be run on energy stored in a rotating flywheel, with an electric motor getting the flywheel up to its top speed of $200\pi\text{ rad/s}$. One such flywheel is a solid, uniform cylinder with a mass of 500 kg and a radius of 1.0 m . (a) What is the kinetic energy of the flywheel after charging? (4)
- (b) If the truck uses an average power of 8.0 kW , for how many minutes can it operate between chargings?

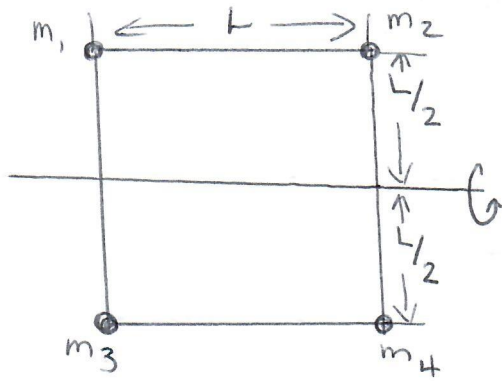
Lorries kan op die energie gestoor in 'n roterende vliegwiel loop, saam met 'n elektriese motor wat die vliegwiel tot op maksimum spoed kry van $200\pi\text{ rad/s}$. Een so vliegwiel is 'n soliede, uniforme silinder met 'n massa van 500 kg en 'n radius van 1.0 m . (a) Wat is die kinetiese energie van die vliegwiel nadat dit gelaaai is?

(b) Indien die lorie 'n gemiddelde drywing van 8.0 kW gebruik, hoe lank kan dit loop voordat dit weer gelaaai moet word?

(6)

Tutorial Test 10 - 2.

1.



$$\begin{aligned} I &= I_1 + I_2 + I_3 + I_4 \\ &= m_1 \left(\frac{L}{2} \right)^2 + m_2 \left(\frac{L}{2} \right)^2 + m_3 \left(\frac{L}{2} \right)^2 + m_4 \left(\frac{L}{2} \right)^2 \\ &= 4m \left(\frac{L^2}{4} \right) \\ &= mL^2 \\ &= (0.50 \text{ kg})(2.0 \text{ m})^2 = 2.0 \text{ kg} \cdot \text{m}^2 \end{aligned}$$

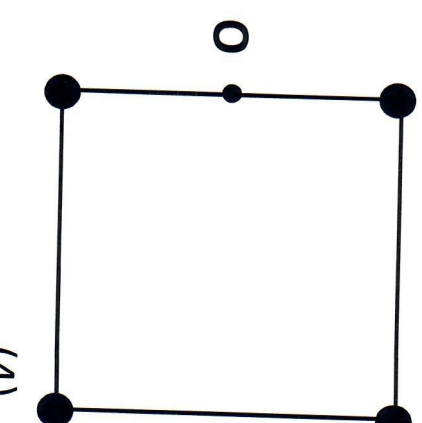
2.a) $K = \frac{1}{2} I \omega^2$ For cylinder, $I = \frac{1}{2} MR^2$

$$\begin{aligned} &= \frac{1}{2} \left(\frac{1}{2} MR^2 \right) \omega^2 \\ &= \frac{1}{4} MR^2 \omega^2 = \frac{1}{4} (500 \text{ kg})(1.0 \text{ m})^2 (200\pi \text{ rad/s})^2 \\ &= 4.9 \times 10^7 \text{ J} \end{aligned}$$

b) $P = \frac{W}{t} = \frac{\Delta K}{\Delta t}$

$$\begin{aligned} \Delta t &= \frac{\Delta K}{P} \\ &= \frac{4.9 \times 10^7 \text{ J}}{8.0 \times 10^3 \text{ W}} \\ &= 6.2 \times 10^3 \text{ s} \end{aligned}$$

1. Four identical particles of mass 0.50 kg each are placed at the vertices of a $2.0\text{ m} \times 2.0\text{ m}$ square and held there by four massless rods, which form the sides of the square. What is the rotational inertia of this rigid body about an axis that passes through the midpoint of one of the sides and is perpendicular to the plane of the square, marked O?



(4)

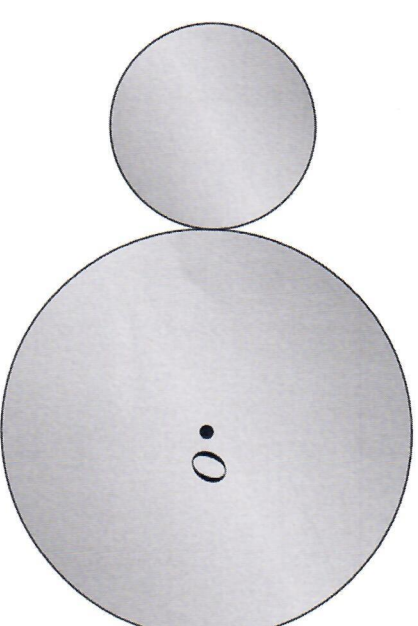
2. Trucks can be run on energy stored in a rotating flywheel, with an electric motor getting the flywheel up to its top speed of 200 rad/s . One such flywheel is a solid, uniform cylinder with a mass of 600 kg and a radius of 1.0 m . (a) What is the kinetic energy of the flywheel after charging?
- (b) If the truck uses an average power of 8.0 kW , for how many minutes can it operate between chargings?

Lorries kan op die energie gestoor in 'n roterende vliegwiel loop, saam met 'n elektriese motor wat die vliegwiel tot op maksimum spoed kry van $200\pi\text{ rad/s}$. Een so vliegwiel is 'n soliede, uniforme silinder met 'n massa van 600 kg en 'n radius van 1.0 m . (a) Wat is die kinetiese energie van die vliegwiel nadat dit gelaai is?

(b) Indien die lorie ? gemiddelde drywing van 8.0 kW gebruik, hoe lank kan dit loop voordat dit weer gelaai moet word?

(6)

1. In the figure, a small disk of radius $r = 2.00$ cm has been glued to the edge of a larger disk of radius $R = 4.00$ cm so that the disks lie in the same plane. The disks can be rotated around a perpendicular axis through point O at the center of the larger disk. The disks both have a uniform density (mass per unit volume) of 1.40×10^3 kg/m³ and a uniform thickness of 5.00 mm. What is the rotational inertia of the assembly about the rotation axis through O? Hint:

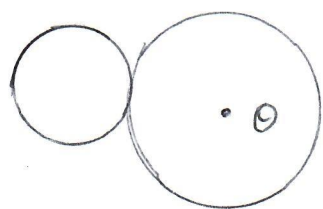


$$\rho = \frac{\text{Mass}}{\text{Volume}}.$$

Die figuur dui 'n klein skyf met radius $r = 2.00$ cm wat vasgeplak is aan die kant van 'n groter skyf met radius $R = 4.00$ cm sodat die twee skywe in dieselfde vlak lê. Die skywe roteer om 'n as loodreg op die vlak deur die punt O by die middelpunt van die groter skyf. Beide skywe het 'n uniforme digtheid van 1.4×10^3 kg/m³ en uniforme dikte van 5.00 mm. Wat is die traagheidsmoment van die stelsel om die rotasie as deur O?? Wenk:

$$\rho = \frac{\text{massa}}{\text{volume}}.$$

Tutorial 10 - 4



$$I_{\text{disk}} = \frac{1}{2} MR^2$$

$$\begin{aligned} I &= I_1 + I_2 \\ &= \frac{1}{2} MR^2 + \left(\frac{1}{2} mr^2 + m(r+R)^2 \right) \end{aligned}$$

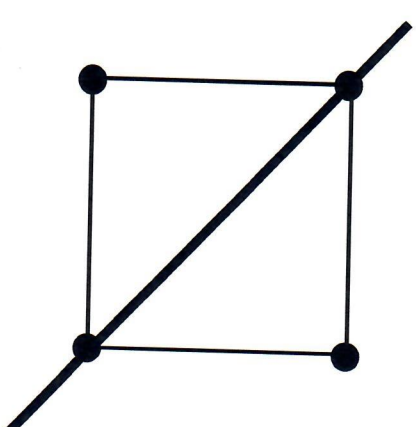
$$m_1 = \rho V = \rho (\pi r^2) t, \quad t = \text{thickness}$$

$$\begin{aligned} m_1 &= 1.40 \times 10^3 \text{ kg/m}^3 \times \pi (2.00 \times 10^{-2} \text{ m})^2 (5 \times 10^{-3} \text{ m}) \\ &= 8.80 \times 10^{-3} \text{ kg} \end{aligned}$$

$$\begin{aligned} M &= 1.40 \times 10^3 \text{ kg/m}^3 \times \pi (4.00 \times 10^{-2} \text{ m})^2 (5 \times 10^{-3} \text{ m}) \\ &= 0.0352 \text{ kg} = 3.52 \times 10^{-2} \text{ kg} \end{aligned}$$

$$\begin{aligned} I &= \frac{1}{2} (3.52 \times 10^{-2} \text{ kg}) (4.00 \times 10^{-2} \text{ m})^2 + \frac{1}{2} (8.80 \times 10^{-3} \text{ kg}) \times \\ &\quad (2.00 \times 10^{-2} \text{ m})^2 + (8.80 \times 10^{-3} \text{ kg}) (6 \times 10^{-2} \text{ m})^2 \\ &= 6.16 \times 10^{-5} \text{ kg} \cdot \text{m}^2 \end{aligned}$$

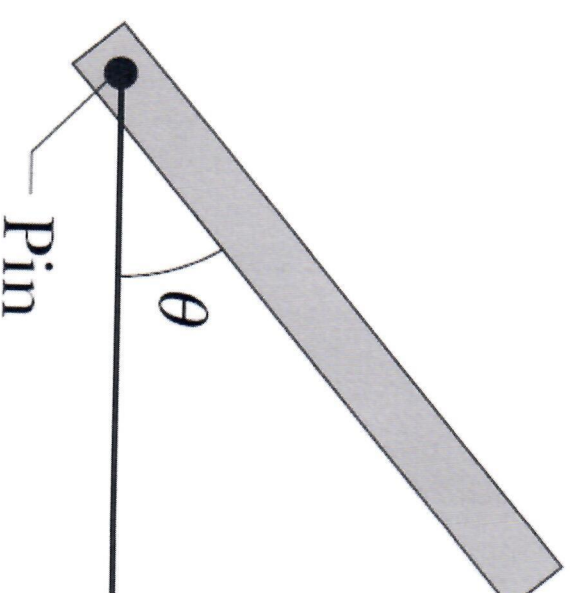
1. Four identical particles of mass 0.50 kg each are placed at the vertices of a $2.0\text{ m} \times 2.0\text{ m}$ square and held there by four massless rods, which form the sides of the square. What is the rotational inertia of this rigid body about an axis that lies in the plane of the square and passes through two diagonally opposite particles, as shown in the figure?



- Vier identiese deeltjies elk met massa 0.50 kg word op die hoeke van 'n $2.0\text{ m} \times 2.0\text{ m}$ vierkant geplaas. Hulle word daar in plek gehou deur vier massalose stawe wat die sykante van die vierkant vorm. Wat is die traagheidsmoment van die rigiede liggaam om 'n as wat in die vlak van die vierkant lê en deur twee deeltjies wat diagonaal teenoormekaar lê soos aangedui in die figuur?*

2. The thin uniform rod in the figure has length 2.0 m and can pivot about a horizontal, frictionless pin through one end. It is released from rest at angle $\theta = 40^\circ$ above the horizontal. Use the principle of conservation of energy to determine the angular speed of the rod as it passes through the horizontal position.

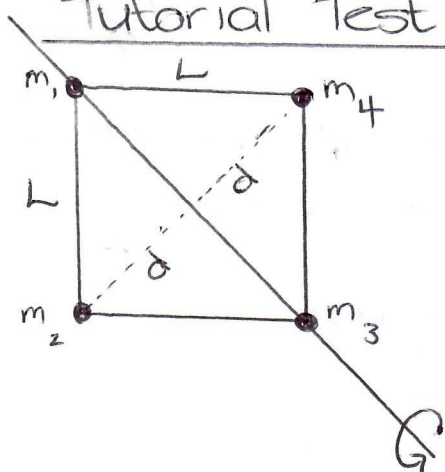
- Die uniforme staaf in die figuur het 'n lengte 2.0 m en kan roteer om 'n horisontale, wrywinglose pen wat deur een kant. Dit word gelos uit rus teen 'n hoek $\theta = 40^\circ$ bo die horisontaal. Gebruik behoud van energie en bepaal die hoeksnelheid van die staaf soos dit deur die horisontale posisie beweeg.*



(5)

Tutorial Test 10-5

1.



$$2d = \sqrt{L^2 + L^2}$$

$$= \sqrt{2L^2}$$

$$d = \frac{1}{2} \sqrt{2L^2}$$

$$= \sqrt{\frac{1}{2} L^2}$$

$$\text{or } L^2 = d^2 + d^2 = 2d^2$$

$$d = \sqrt{\frac{1}{2} L^2}$$

$$I = I_1 + I_2 + I_3 + I_4$$

$$= 0 + md^2 + 0 + md^2$$

$$= 2m \left(\sqrt{\frac{1}{2} L^2} \right)^2 = 2m \left(\frac{1}{2} L^2 \right)$$

$$= 2(0.5 \text{ kg}) \left(\frac{1}{2} \right) (2.0 \text{ m})^2$$

$$= 2.0 \text{ kg} \cdot \text{m}^2$$

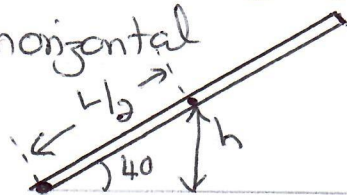
2. com of rod is at the centre.

Initially it is at a height h above the horizontal. Take $U_g = 0$ along horizontal

$$h = \frac{L}{2} \sin 40^\circ$$

The potential energy of rod is

$$U_g = mgh$$



As rod passes through horizontal $K = \frac{1}{2} I \omega^2$

Using conservation of energy:

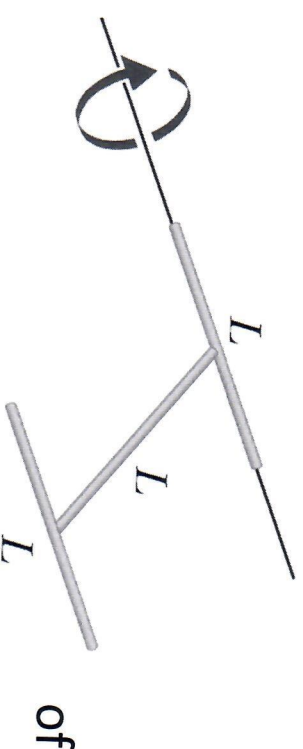
$$U_i + K_i = U_f + K_f$$

$$mg \left(\frac{L}{2} \sin 40^\circ \right) + 0 = 0 + \frac{1}{2} I \omega^2 = \frac{1}{2} \left(\frac{1}{3} m L^2 \right) \omega^2$$

$$\omega = \sqrt{\frac{3g \sin 40^\circ}{L}} = 3.07 \text{ rad/s}$$

A rigid body is made of three identical thin rods, each with length $L = 0.600\text{ m}$, fastened together in the form of a letter H (see Figure). The body is free to rotate about a horizontal axis that runs along the length of one of the legs of the H. The body is allowed to fall from rest from a position in which the plane of the H is horizontal.

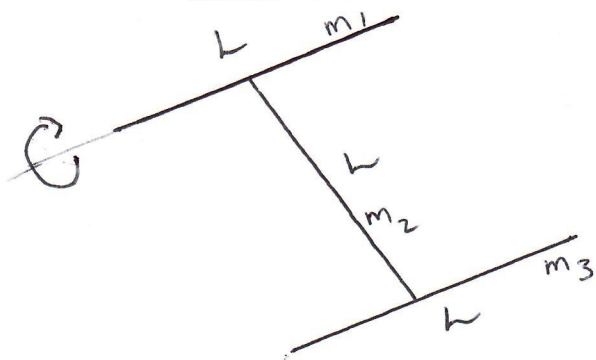
- (a) Find the rotational inertia of the body about the axis indicated. (4)
- (b) What is the change in the potential energy of the body when it falls from the horizontal position to when the plane of the H is vertical? (2)
- (c) Using energy methods determine the angular speed when the plane of the H is vertical? (4)



’n Rigiëde liggaam word uit drie dun stafies gemaak, elk van lengte $L = 0.600\text{ m}$, en hulle word so vasgemaak dat hulle ’n letter H vorm (Sien figuur). Die liggaam is vry om te roteer om ’n horisontale as wat deur een van die bene van die H lê. Die liggaam word toegelaat om uit rus te val van die oorspronklike horisontale posisie.

- (a) Bepaal die traagheidsmoment van die liggaam om die as wat aangedui is. (4)
- (b) Wat is die verandering in die potensiële energie van die liggaam wanneer dit van die horisontale posisie van die H na die vertikale posisie van die H val? (2)
- (c) Gebruik energie metodes om die hoek snelheid te bepaal wanneer die H vertikaal lê. (4)

Tutorial Test 10-6



$$m_1 = m_2 = m_3 = m$$

Total mass: $3m$

$$\begin{aligned} \text{a) } I &= I_1 + I_2 + I_3 \\ &= 0 + \frac{1}{3}mL^2 + (mL^2) \\ &= \frac{4}{3}mL^2 \end{aligned}$$

$$\text{b) } \Delta U = U_f - U_i = mg(\Delta h)$$

com of H is at the centre of the crossbar.

$\therefore$ com drops (from horizontal plane to vertical plane) by a vertical distance of $-\frac{L}{2} = \Delta h$

$$\Delta U = -3mg\left(\frac{L}{2}\right)$$

c) Conservation of Energy

Take $U_g = 0$ in horizontal position ($y = 0$)

$$\begin{aligned} K_i + U_i &= K_f + U_f \\ 0 + 0 &= \frac{1}{2}I\omega^2 + 3mg\left(-\frac{L}{2}\right) \end{aligned}$$

$$\omega^2 = \frac{+mgL/2}{\frac{1}{2}I} = \frac{3mgL}{\frac{4}{3}mL^2} = \frac{9}{4} \frac{g}{L}$$

$$\omega = \sqrt{\frac{9}{4} \frac{9.8 \text{ m/s}^2}{0.6 \text{ m}}} = 6.06 \text{ rad/s}$$

$$= 6.1 \text{ rad/s}$$